

AI61002 : Deep Learning Foundations and Applications

Spring 2026

Assignment 2: Convolutional Neural Networks

(20 Marks)

Instructions: Complete the tasks listed below, ensuring you follow the specified instructions for each. Provide step-by-step outputs in the Jupyter Notebook (.ipynb file). All relevant code snippets, visualizations, and explanations must be included within the notebook. Submit the required files as per the submission guidelines provided.

Question 1 - 1 mark

Load the EUROSAT dataset from Torchvision:

<https://docs.pytorch.org/vision/0.20/generated/torchvision.datasets.EuroSAT.html> and show the class distribution of the dataset.

Question 2 - 1 mark

Split the dataset into training, validation and test sets (70%-15%-15%)

Question 3 - 2 marks

Create the data loaders which perform data augmentation using flipping, rotation and random crop+resize.

Question 4 - 3 marks

Using PyTorch, design a fully convolutional neural network (with GAP and 1x1 convolution) with residual connections to perform classification on this dataset

Question 5 - 2 marks

Train the CNN you designed using the Adam optimizer to minimize a suitable loss function. Show the variation of the training and validation losses with training progress.

Question 6 - 2 marks

Report the performance of the trained CNN model on the test set in terms of precision and recall. Also, show the confusion matrix.

Question 7 - 4 marks

Implement a DeepFool attack on the trained model to misclassify the samples using the following algorithm from

Algorithm 2 DeepFool: multi-class case

```
1: input: Image  $\mathbf{x}$ , classifier  $f$ .
2: output: Perturbation  $\hat{\mathbf{r}}$ .
3:
4: Initialize  $\mathbf{x}_0 \leftarrow \mathbf{x}, i \leftarrow 0$ .
5: while  $\hat{k}(\mathbf{x}_i) = \hat{k}(\mathbf{x}_0)$  do
6:   for  $k \neq \hat{k}(\mathbf{x}_0)$  do
7:      $\mathbf{w}'_k \leftarrow \nabla f_k(\mathbf{x}_i) - \nabla f_{\hat{k}(\mathbf{x}_0)}(\mathbf{x}_i)$ 
8:      $f'_k \leftarrow f_k(\mathbf{x}_i) - f_{\hat{k}(\mathbf{x}_0)}(\mathbf{x}_i)$ 
9:   end for
10:   $\hat{l} \leftarrow \arg \min_{k \neq \hat{k}(\mathbf{x}_0)} \frac{|f'_k|}{\|\mathbf{w}'_k\|_2}$ 
11:   $\mathbf{r}_i \leftarrow \frac{|f'_l|}{\|\mathbf{w}'_l\|_2} \mathbf{w}'_l$ 
12:   $\mathbf{x}_{i+1} \leftarrow \mathbf{x}_i + \mathbf{r}_i$ 
13:   $i \leftarrow i + 1$ 
14: end while
15: return  $\hat{\mathbf{r}} = \sum_i \mathbf{r}_i$ 
```

Question 8 - 2 marks

Perform the adversarial attack designed above and report its success rate (note success rate would be the fraction of samples that were initially correctly classified by the trained model but misclassified with their adversarial versions)

Question 9 - 3 marks

Use the torchattack library from <https://pypi.org/project/torchattack/> to perform the same attack and compare its success with that attained by your implementation:

Submission Guidelines

1. The content that you submit must be your individual work.
2. Submit your code in .py as well as in .ipynb file format. Both these file submissions are required to receive credit for this assignment.
3. Ensure your code is well-commented and easy to follow. You can write your answers and explanations using text cells in the jupyter notebook files wherever required.

4. The files should be named as “(roll_number)_assignment_2”. For example, if your roll number is 23AI91R01, the code file names will be 23AI91R01_assignment_2.py and 23AI91R01_assignment_2.ipynb. You should place all these files within a single .zip file (do not upload a .rar file) and upload it to Moodle as 23AI91R01_assignment_2.zip. The zip file should only contain the .py and .ipynb files, and nothing else.
5. All submissions must be made through Moodle before the deadline. The submission portal will close at the specified time, and submissions via email would not be accepted.
6. The .ipynb file acts as your assignment report in addition to the implementation. Therefore, ensure that the .ipynb file is clear and easy to assess. To discourage plagiarism, the .py file is used to check for plagiarism with very strict deduction criteria. Anyone trying to bypass the plagiarism check with means such as gibberish text inside the code will also experience harsh deduction.
7. The primary TA for assignment 2 is Raj Krishan Ghosh (rajkrishanghosh@kgpian.iitkgp.ac.in). In case you have any query regarding the assignment, you can email the TA. Please do not call.